

Md. Touhid Hasan Nirob

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 Google Scholar

Location: Islamic University, Kushtia-7003, Bangladesh

 LinkedIn Profile

CAREER OBJECTIVE

Aspiring researcher with a strong interest in Artificial Intelligence, Medical Imaging, Few-Shot Learning, Federated Learning, and Embedded Systems. Committed to advancing intelligent solutions for healthcare and next-generation engineering applications.

EDUCATION

- B.Sc Engineering in Electrical and Electronic Engineering**
Islamic University, Bangladesh 2022 – Present
CGPA: 3.83/4.00 (upto 7th semester)
- Higher Secondary School Certificate (H.S.C)**
Nawabganj Govt. College, Chapainawabganj, Rajshahi Year: 2020
GPA: 5.00/5.00
- Secondary School Certificate (S.S.C)**
Bhitorbanda J.D. Academy, Nageswary, Kurigram Year: 2018
GPA: 5.00/5.00

PUBLICATIONS & RESEARCH

JOURNAL ARTICLES:

- Md. Touhid Hasan Nirob**, Md. Fattah Al Fahim, Md. Mahfuz Rahman Anik, Md. Mahfuz Ahmed, Md. Maruf Hossain, Md. Khairul Islam, M. Jalal Uddin, “NEURO-PULM: A Deployable Hybrid XceptionNet-ViT Model with Explainable AI for Tuberculosis and Pneumonia Classification from Chest Radiographs,” *Submitted to Computer Methods and Programs in Biomedicine*.
- Md. Mahfuz Ahmed, Md. Maruf Hossain, Md. Rakibul Hasan Rakib, Ronok Hashan, **Md. Touhid Hasan Nirob**, Md. Khairul Islam, “A hybrid swin transformer–BiLSTM framework and ensemble learning for multimodal brain stroke detection and risk prediction,” *Computers in Biology and Medicine*, Volume 187, 2026. DOI: 10.1016/j.combiomed.2026.111518

CONFERENCE PROCEEDINGS:

- Md. Touhid Hasan Nirob**, Md. Mahfuz Rahman, Md. Fattah Al Fahim, Md. Mahfuz Ahmed, Md. Rakibul Hasan Rakib, Md. Maruf Hossain, Md. Khairul Islam, Md. Jalal Uddin, “A Lightweight Deployment-Optimized Deep Learning Framework for Multi-Class Pulmonary Disease Diagnosis from CXRs with XAI Integration,” *IEEE QPAIN 2026*. DOI: 10.1109/QPAIN69676.2026.11546323
- Md. Fattah Al Fahim, **Md. Touhid Hasan Nirob**, Md. Mahfuz Rahman Anik, Md. Mahfuz Ahmed, Md. Rakibul Hasan, Md. Arman Hossain Joy, Md. Khairul Islam, Md. Jalal Uddin, “A Privacy-Preserving Customized CNN-Federated Learning Framework with XAI for Lung Cancer Detection and Classification Using CT Images,” *IEEE QPAIN 2026*. DOI: 10.1109/QPAIN69676.2026.11546098
- Md. Mahfuz Rahman Anik, **Md. Touhid Hasan Nirob**, Md. Fattah Al Fahim, Md. Mahfuz Ahmed, Jannatul Fardous, Sabbir Hossain, Md. Khairul Islam, Md. Jalal Uddin, “ConvEffNET: A Hybrid Deep Learning Framework with Explainable AI for Multi-Class Heart Diseases Classification Using Echocardiogram Images,” *IEEE QPAIN 2026*. DOI: 10.1109/QPAIN69676.2026.11545984

- Md. Rakibul Hasan Rakib, Md. Mahfuz Ahmed, **Md. Touhid Hasan Nirob**, Amimul A. Hasan Rofik, Md. Mahfuz Rahman Anik, “A Hybrid Swin Transformer–VGG19 Framework Enhanced with Explainable Artificial Intelligence for Accurate Brain Tumor Detection and Classification Using MRI Data,” 28th International Conference on Computer and Information Technology (ICCIT 2025), IEEE, Cox’s Bazar, Bangladesh, 2025. DOI: 10.1109/ICCIT68739.2025.11491417

SYMPOSIUM PRESENTATIONS

- **Md. Touhid Hasan Nirob**, Md. Fattah Al Fahim, Md. Mahfuz Rahman Anik, “Deep Learning-Based Multi-Class Lung Disease Classification Using Chest X-ray Images with High Precision,” *IEEE CS BDC Summer Symposium 2025*, Paper ID: 396.
- Md. Maruf Hossain, Md. Mahfuz Ahmed, **Md. Touhid Hasan Nirob**, Md. Mahfuz Rahman Anik, Md. Sipon Miah, Md. Khairul Islam, “Multi-Modal Federated Learning Framework for Automated Lung Tuberculosis Detection and Classification Using X-Ray Imaging,” *IEEE CS BDC Summer Symposium 2025*, Paper ID: 315.

PERSONAL PROJECTS

- **Bidirectional Medical Image Translation (CT ↔ MRI) using Advanced CycleGAN** Link
Healthcare AI | Deep Learning | Medical Imaging | Generative AI

- Developed an advanced CycleGAN-based framework for bidirectional medical image translation (CT ↔ MRI) using 1,568 CT and 1,570 MRI unpaired images. Designed dual generators and PatchGAN discriminators with cycle consistency, identity loss, and SSIM-based optimization to preserve anatomical structures and improve cross-modality image synthesis for AI-assisted clinical workflows.

Tools & Technologies Used: Python, PyTorch, CycleGAN, Deep Learning, Medical Imaging, OpenCV, SSIM, Image-to-Image Translation, NVIDIA GPU
- **Federated Learning-Based Tuberculosis Detection from Chest X-ray Images** Link
Healthcare AI | Federated Learning | Medical Imaging

- Developed a federated learning framework that enables multiple healthcare institutions to collaboratively train deep learning models without sharing sensitive patient data. Implemented secure model aggregation and decentralized training to preserve privacy while maintaining high predictive performance for medical diagnosis applications.

Tools & Technologies Used: Python, PyTorch, Federated Learning (FedAvg), Flower, CNN, Medical Imaging, OpenCV, NVIDIA GPU
- **NEURO-PULM: Deep Learning based Multi-Disease Detection from Chest X-rays** Link
Team FutureFrame | Deep Learning, Medical Imaging

- Developed a deep learning model (ResNet + LSTM hybrid) to detect pulmonary diseases, including TB and Pneumonia, with up to 97% Accuracy and AUC of 0.97. Designed with explainable AI (Grad-CAM) for transparent diagnosis and deployed via web and mobile apps for real-time inference.

Tools & Technologies Used: Python, PyTorch, CNN, LSTM, Grad-CAM, Streamlit, NVIDIA T4 GPU, OpenCV
- **AI-Powered Tuberculosis Detection System** Link
Machine Learning for Healthcare

- Developed a machine learning model capable of detecting pulmonary tuberculosis from chest X-ray images with 90% accuracy. This project aimed to support early TB diagnosis in low-resource settings, contributing to AI-driven medical diagnostics.

Tools & Technologies Used: Python, TensorFlow, CNN, Scikit-learn, OpenCV, Jupyter Notebook
- **Locker Security Enhancer: Ultrasonic Shield Security System with Biometric Access, Real-time Alert and Photo Evidence.** Link
Embedded Systems | IoT-enabled Security Device

- Led the development of a portable, IoT-integrated locker security system with four protection layers: ultrasonic intrusion detection, fingerprint authentication, real-time GSM alerts, and ESP32-CAM-based photo capture and instantaneous upload to Google Drive.

Tools & Technologies Used: Arduino UNO, ESP32-CAM, GSM Module (SIM800L), Ultrasonic Sensor, PIR Sensor, Fingerprint Sensor, C/C++, Google Drive API

PROFESSIONAL EXPERIENCE

- Bio-Imaging Research Laboratory** December 2024 – Present
Islamic University, Kushtia-7003, Bangladesh
Research Assistant
- Smart Sensor & Microactuators Research & Training (SMaRT) Lab** October 2025 – Present
Islamic University, Kushtia-7003, Bangladesh.
Research Assistant
- IoT Lab** June 2022 – Sept.2023
Electrical & Electronic Engineering Department
Islamic University, Kushtia-7003, Bangladesh.
Project Researcher
- Private Academic Tutor - Physics & Mathematics** May 2023 – Present
Self-Employed

TECHNICAL SKILLS

- Programming:** Python, MATLAB, C, Arduino Sketch
- AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras
- Libraries:** OpenCV, NumPy, Pandas, SciPy, Matplotlib, Seaborn
- Research Tools:** LaTeX, Overleaf, Draw.io, Canva, Anaconda, Kaggle, Jupyter Notebook, Google Colab
- Version Control:** Git, GitHub
- Simulation Software:** MATLAB/Simulink, Proteus
- Embedded Systems & IoT:** Arduino, ESP32, ESP8266, GSM Modules, Sensors, Microcontrollers
- Research Domains:** Medical Image Analysis, Deep Learning, Federated Learning, Few-Shot Learning, Computer Vision, Explainable AI, IoT & Embedded Systems

HONORS & AWARDS:

- Appointed as Parallel Session Moderator (AI & Machine Learning Track) and Session Moderator** October, 2025
IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025)
- Talent Pool Scholarship (Top Merit Category): *B.Sc. 3rd Year Academic Award*** 2nd Position
Department of Electrical and Electronic Engineering, Islamic University, Bangladesh CGPA: **3.86/4.00**
- Talent Pool Scholarship (Top Merit Category): *B.Sc. 2nd Year Academic Award*** 2nd Position
Department of Electrical and Electronic Engineering, Islamic University, Bangladesh CGPA: **3.82/4.00**
- Talent Pool Scholarship (Top Merit Category): *B.Sc. 1st Year Academic Award*** 1st Position
Department of Electrical and Electronic Engineering, Islamic University, Bangladesh CGPA: **3.81/4.00**
- National STEAM Olympiad: *Division Round Winner*** May, 2024
Engineering Project Segment

CERTIFICATIONS & COURSES:

- Completed a 21-day Industrial Training program at TICI**
Instrumentation, Control Systems (PLC, SCADA, DCS), and Electrical Engineering. February, 2026
Grade: A+ | Duration: 3 weeks
- Completion of Machine Learning Course**
An online course from Study Mart July, 2024
- Completion of Python Course**
An online course from Coursera June, 2023
- National Skill Standard Basic Certificate (Computer Office Application)**
Bangladesh Technical Education Board (BTEB) Jan 2018– Jun 2018
Grade: A+ | Duration: 6 Months (360 Hours)
- COVID-19 Awareness and Safety Training (Healthcare Focus)**
Muktopaath (Govt. e-learning platform) – MC-B518899S796928G236T May, 2020
- COVID-19 Health Worker Training Program**
Jeeon & Muktopaath Collaborative Program – MC-N518899N819142T242Q August, 2020

VOLUNTEERING ACTIVITIES

- IEEE IU Student Branch**
 - Vice-Chair (Technical) July, 2026 – Present
 - Joint Secretary September, 2025 – July, 2026
 - Membership Development Coordinator June, 2024 – September, 2025
 - Member March, 2023 – June, 2024
- Kurigram Student Welfare Association**
Islamic University, Bangladesh
 - Vice President July, 2026 – Present
 - Joint Secretary December, 2024 – July, 2026
 - Member August, 2022 – December 2024
- Islamic University Science Club**
 - Executive October, 2022 – May 2024